(1.) Course Title: INFPHD470 AI Infused Data Science Solutions in ERP Systems with SAP Case Studies	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: (50-50%) (credit%)	
Type of Class: lecture and **number of hours**: **28** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): midterm report **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: theme elaborations	
Course Placement in Curriculum (Semester Number): 2	
Prerequisites (*if any*): -	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

The goal of the subject is to introduce built-in AI models and technologies in commercial ERP Systems and their possible enhancements by custom AI services developments. Efficiency of different AI models are evaluated, their advantageous and disadvantageous features are demonstrated by case studies. Dependency of appropriate AI solutions on ERP System parameters, such as the architecture of the information system, applied novel technologies, current versions of the ERP software are investigated. Applications of open-source technologies to commercial ERP solutions extend the interoperability between open-source and commercial technologies.

The course is concerned with introducing the basic concepts in data science through real-world examples from various application areas with special focus on telecommunication, agriculture and enterprise management. Data science techniques will be dealt with in a black-box manner, i.e. the focus is to show what is on their input and output as well as how to interpret the resulting models. During the course, students work on a semester project which quality will determine the final grade.

The main modules/topics of the course include the following:

1. Data Science and Artificial Intelligence in Enterprise Resource Systems
2. ERP Case Studies and Technology Platform
3. Individual Project Selecrions
4. Research Goals, Approaches
5. Scientific and Project Work

Presentation of individual project and scientific achievements

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

1. Pradeep K. Suri: AI-powered Enterprise Resource Planning. ISBN: 978-93-5819-249-0. BlueRoseOne, 2023.

2. Chris Kutt: ERP with Confidence: The Ultimate Guide for Middle Market Professionals Navigating the ERP Journey. ISBN-13 : 979-8218260255. Yard-Hard Multimedia Company LLC, 2023

3. Liam Caton: A Complete Guide to AI for Business: Boost Productivity, Enhance Customer Experience, and Improve Decision Making. ISBN-13: 979-8867017026. Independently published, 2023

4. Dr. Amol S. Kalgaonkar: ERP Revolution: Transforming the Future of Business: Unleashing the Path to Business Excellence and Competitive Advantage for the Digital Age (Kindle Edition). Book Saga Publications, 2023

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Dr. Tamás Gábor Orosz, habil. associate professor, Ph.D.**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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